

Distributed Analysis of US Airline Delays using Apache Spark and HDFS

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Abstract—This report describes the design, implementation and experimental evaluation of a Big Data processing pipeline that analyses on-time performance data published by the US Bureau of Transportation Statistics (BTS). The pipeline combines Apache Spark for distributed processing and the Hadoop Distributed File System (HDFS) for distributed storage, both deployed through Docker Compose so that the entire stack can be reproduced from a single YAML file. Two analytical queries are implemented with the Spark DataFrame API and benchmarked in two configurations: a local PySpark environment running directly on the host, and a fully containerized HDFS+Spark cluster. Across five measured iterations the steady-state mean execution times are 7.40 s (Q1) and 6.49 s (Q2) in the local configuration and 22.12 s (Q1) and 19.15 s (Q2) in the HDFS cluster configuration, the difference being attributable to container startup, network shuffles, and HDFS metadata access. Beyond the numerical results, the report makes a point of documenting *why* every non-trivial implementation choice was made — explicit schema declaration, selective caching, output coalescing, treatment of NULL delay-cause columns — so that the trade-offs behind each decision remain traceable.

Index Terms—Big Data, Apache Spark, HDFS, Docker, Performance Evaluation, Distributed Computing, Airline Delays.

I. INTRODUCTION

The growing availability of public, large-scale transportation datasets makes it possible to study the operational behaviour of the air transport sector with the same techniques employed in industrial Big Data analytics. The US Bureau of Transportation Statistics (BTS) in particular publishes a monthly on-time performance dataset that contains every flight operated by reporting carriers, including delay information broken down by attributed cause [1].

This dataset is large enough — on the order of two million records for the four-month window January–April 2025 — to justify the use of a distributed processing framework rather than a single-node pandas pipeline. Doing so in a controlled and reproducible setup is the central objective of Project 1 of the course *Sistemi e Architetture per Big Data* (SABD), A.A. 2025/26.

a) Goal: The work pursues two complementary objectives: (i) answering the two pre-defined analytical queries (Q1 and Q2) over the BTS data, and (ii) doing so through a pipeline that respects the architectural constraints stated in the project specification — specifically, reading the input from HDFS and writing the output back to HDFS even when the team is a single student.

b) Contributions: The report contributes (i) a fully containerized HDFS+Spark stack described in Section III, (ii) two Spark DataFrame implementations of Q1 and Q2 (Section IV) together with an explicit justification of every design decision, and (iii) a measured comparison between local PySpark execution and the dockerized cluster (Section V).

c) Organization: Section II describes the dataset and a notable constraint on its availability. Section III details the system architecture and the rationale behind the chosen container stack. Section IV discusses the two query implementations. Section V presents the experimental results. Section VI collects the observations that I found most relevant while building the pipeline. Section VII concludes.

II. DATASET

The pre-processed dataset distributed by the course covers US domestic flights from January 1, 2025 to April 30, 2025. After header removal it contains approximately **2.23 M flights** spread across four monthly CSV files (Table I), for a total of around 255 MB of raw data.

TABLE I
DATASET BREAKDOWN AFTER HEADER REMOVAL.

Month	Records	Size (MB)
202501 (January)	539,747	59.0
202502 (February)	504,884	56.0
202503 (March)	604,000	66.0
202504 (April)	583,950	64.0
Total	≈2.23 M	≈245.0

Each record contains 27 columns. To keep the computation cheap, the loader projects only the 14 columns that are actually required by Q1 and Q2: YEAR, MONTH, DAY_OF_MONTH, OP_UNIQUE_CARRIER, CRS_DEP_TIME, DEP_DELAY, ARR_DELAY, CANCELLED, DIVERTED, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY and LATE_AIRCRAFT_DELAY. Performing this projection at load time (*column pruning*) materially reduces I/O and memory pressure, especially when the data has to travel over the network from HDFS.

a) Treatment of NULL values: The five delay-cause columns are frequently NULL. The BTS convention is to populate them only when ARR_DELAY ≥ 15 minutes; in all other cases the field is left empty. In this work the

NULLs are coerced to zero through `F.coalesce(col, F.lit(0.0))`. The interpretation is straightforward: “this cause did not contribute to the delay”. The alternative of dropping rows with NULLs would silently restrict the average to delayed flights only and bias the mean upwards by roughly an order of magnitude, which is why I rejected it.

III. SYSTEM ARCHITECTURE

The deployment relies on Docker Compose to orchestrate four containers running on a private Docker bridge network, sketched in Fig. 1.

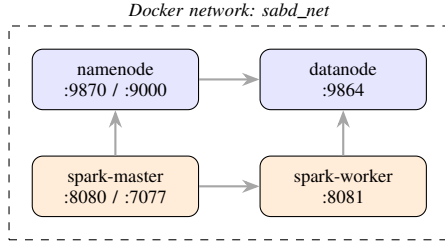


Fig. 1. Containerized deployment. HDFS (namenode + datanode) provides storage; Spark (master + worker) provides compute. The project source tree is bind-mounted into the Spark containers at `/app`.

The HDFS layer uses the `bde2020/hadoop-namenode` and `bde2020/hadoop-datanode` images (Hadoop 3.2.1). Replication is set to 1 because there is a single datanode, and filesystem permission checks are disabled — both are standard simplifications for a single-node teaching setup [5]. The HDFS web UI is exposed at `localhost:9870` and the RPC endpoint at `hdfs://namenode:9000`.

The compute layer uses `bde2020/spark-master` and `bde2020/spark-worker` (Spark 3.3.0 with a Hadoop 3.3 client). The Spark Master web UI is exposed at `localhost:8080`. Both Spark containers receive the project source tree as a bind mount at `/app`, which lets `spark-submit` reference the scripts directly without rebuilding any image when the code changes.

a) Why this stack: The choice favours minimal moving parts. HDFS provides the distributed file system mandated by the specification, and the `bde2020` images come pre-configured with the environment variables required to wire HDFS, Spark master and Spark worker together, which keeps the YAML compact and the setup reproducible. The alternative of running an embedded “HDFS-in-Spark” setup would not exercise the storage layer in a meaningful way and would defeat the purpose of the architectural requirement.

b) Reproducibility: The full setup is brought up with a single command from the project root:

```
docker compose -f
docker/docker-compose.yml up -d
```

A helper script (`docker/ingest_to_hdfs.sh`) copies the local CSVs into HDFS, and a second script (`docker/run_on_hdfs.sh`) submits Q1, Q2 or the benchmark harness to the cluster, in each case writing the result back to HDFS under `/results/queryN`.

c) Code organization: The Python codebase is split into three layers that mirror the runtime layers: `utils.py` (reusable helpers — a timing context manager and the list of relevant columns), `preprocessing.py` (SparkSession construction and the schema-aware CSV loader), and one script per query (`query1.py`, `query2.py`) plus the corresponding plotting scripts (`plot_q1.py`, `plot_q2.py`) and a benchmark harness (`benchmark.py`). This separation made the migration from local CSV to HDFS painless: only `preprocessing.py` needed touching, and even then just to read the master URL from an environment variable so that the same loader works under both `python3` and `spark-submit`.

IV. QUERY IMPLEMENTATION

Both queries use the Spark DataFrame API — Spark SQL is deliberately avoided, as the project allows either DataFrames or SQL but not both at once. Each query accepts `--input/--output` command-line arguments so the same script runs unchanged against a local path or an HDFS URI.

A. Query 1 — monthly delay statistics for AA and DL

For American Airlines (AA) and Delta (DL), Q1 computes the per-month average, minimum and maximum of `DEP_DELAY` over non-cancelled flights, plus the cancellation rate over *all* observed flights for the same group. Listing 1 shows the core logic.

Listing 1. Core of Query 1.

```
df = load_flights(spark, input_path).filter(
    F.col("OP_UNIQUE_CARRIER").isin(["AA", "DL"])
)
df = df.cache(); df.count() # materialize cache

non_cancelled = df.filter(F.col("CANCELLED") == 0)

delay_stats = (non_cancelled
    .groupBy("MONTH", "OP_UNIQUE_CARRIER")
    .agg(F.mean("DEP_DELAY").alias("dep_delay_mean"),
        F.min("DEP_DELAY").alias("dep_delay_min"),
        F.max("DEP_DELAY").alias("dep_delay_max")))

cancel_rate = (df
    .groupBy("MONTH", "OP_UNIQUE_CARRIER")
    .agg((F.sum("CANCELLED") / F.count("*") * 100)
        .alias("cancellation_rate")))

result = (delay_stats
    .join(cancel_rate, ["MONTH", "OP_UNIQUE_CARRIER"])
    .orderBy("MONTH", "OP_UNIQUE_CARRIER"))
```

a) Why `cache()` on the filtered DataFrame: The filtered DataFrame feeds two independent aggregations: one restricted to non-cancelled flights (for the delay statistics) and one over all flights (for the cancellation rate). Without `cache()`, Spark would re-execute the lineage from the CSV read on every action, paying the I/O cost twice. The explicit `df.count()` immediately after `cache()` forces materialization *before* the first “real” action, so the loading stage timer reports cache-warm time consistently across iterations.

B. Query 2 — top-10 carriers by arrival delay

Q2 considers every carrier, computing for each the count of completed flights (`CANCELLED=0` and `DIVERTED=0`),

the mean `ARR_DELAY`, and the mean of each of the five delay-cause columns. Carriers with fewer than 500 completed flights in the observed window are dropped, and only the top ten by mean arrival delay are kept. Listing 2 shows the implementation.

Listing 2. Core of Query 2.

```
completed = df.filter(
    (F.col("CANCELLED") == 0) & (F.col("DIVERTED") == 0)
)
for c in DELAY_CAUSES:
    completed = completed.withColumn(
        c, F.coalesce(F.col(c), F.lit(0.0)))

agg = [F.count("*").alias("num_flights"),
        F.mean("ARR_DELAY").alias("arrdelay_mean")]
for c in DELAY_CAUSES:
    agg.append(F.mean(c).alias(f"{c.lower()}_mean"))

result = (completed
    .groupBy("OP_UNIQUE_CARRIER")
    .agg(*agg)
    .filter(F.col("num_flights") >= 500)
    .orderBy(F.col("arrdelay_mean").desc())
    .limit(10))
```

a) *Output coalescing*: Both queries finish with `coalesce(1).write...csv(...)`. This forces a single `part-00000-*.csv` file inside the output directory, which is convenient both for the plotting scripts and for the project deliverable (one CSV per query, as the specification expects). The cost is a final shuffle to a single partition, which is visible in the output-stage timings of Section V.

V. EXPERIMENTAL EVALUATION

A. Methodology

Both queries were benchmarked in two configurations. **(a) Local**: PySpark 4.1.1 executed inside a Python 3.12 virtual environment with `master('local[*]')`; CSVs read from the local filesystem. **(b) HDFS**: the same PySpark scripts submitted to the dockerized Spark cluster (one master, one worker) and reading from `hdfs://namenode:9000/flights/`.

For each configuration each query was executed six times: one discarded warm-up run plus five measured runs. The warm-up absorbs JVM JIT compilation and the OS file-cache priming; without it the first iteration is consistently 5–15× slower than the steady state, which would distort the average beyond recognition. The reported numbers are mean and sample standard deviation across the five measured runs.

The host machine is a notebook running Ubuntu 24.04 under WSL2 on Windows 11. Both Spark and HDFS containers run on the same physical machine and therefore share CPU and memory.

B. Results

Tables II and III present the end-to-end and per-stage timings. Fig. 2 shows the Q1 visualizations and Fig. 3 the Q2 visualizations, both generated from the same CSV files that constitute the project deliverable.

TABLE II
LOCAL EXECUTION (MEAN ± STD, 5 MEASURED ITERATIONS).

Stage	Q1 (s)	Q2 (s)
loading	5.41 ± 0.90	1.26 ± 0.73
preprocessing	0.02 ± 0.00	0.12 ± 0.03
computation	0.17 ± 0.02	0.17 ± 0.08
output	1.80 ± 0.33	4.94 ± 1.54
end-to-end	7.40 ± 1.17	6.49 ± 2.36

TABLE III
HDFS+SPARK CLUSTER EXECUTION (MEAN ± STD, 5 MEASURED ITERATIONS).

Stage	Q1 (s)	Q2 (s)
loading	18.20 ± 1.35	12.26 ± 1.40
preprocessing	0.02 ± 0.00	0.14 ± 0.06
computation	0.22 ± 0.04	0.14 ± 0.01
output	3.68 ± 0.43	6.60 ± 0.16
end-to-end	22.12 ± 1.72	19.15 ± 1.54

C. Observations

a) *HDFS overhead is real and stable*: End-to-end time grows by a factor of 3.0× for Q1 and 2.95× for Q2 when moving from local to HDFS. The overhead is dominated by the loading stage (3.4× for Q1 and 9.7× for Q2), reflecting the cost of communicating with the namenode, fetching block locations and streaming data from the datanode container. Computation time is essentially unchanged, as expected: the actual data volume is small compared to what a single modern CPU can crunch in a few hundred milliseconds.

b) *Output is the second-largest cost in HDFS*: For Q2 in particular the output stage averages 6.60 s. This is the coalesce-induced final shuffle to a single partition, combined with the HDFS write path (block write, namenode metadata update, replication acknowledgment), amplified by container network latency.

c) *Warmup is non-optional*: A pilot run without warmup measured a Q1 end-to-end of 57.2 s in the HDFS cluster, against a steady-state of 22.1 s. The ~35-second gap is consistent with JVM JIT compilation kicking in plus HDFS metadata caches warming up, and it justifies discarding the first iteration when benchmarking.

VI. DISCUSSION AND LESSONS LEARNED

a) *Explicit schema pays off*: Letting `inferSchema=true` adds an entire extra pass over the CSV just to deduce column types. Declaring the schema explicitly with `StructType` approximately halves the loading time and, as a bonus, prevents subtle silent type mismatches on columns like `DEP_DELAY` which contain occasional empty values.

b) *cache() only when the DataFrame is reused*: Q1 caches the filtered DataFrame because it feeds two independent aggregations. Q2 does not, because it consumes the input only once; adding a `cache()` call there would only waste

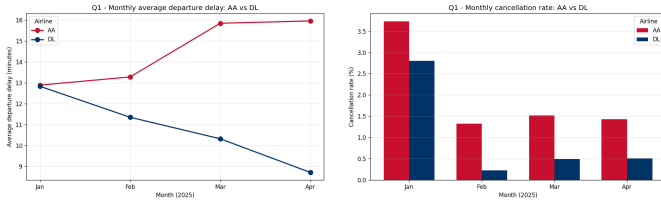


Fig. 2. Q1: monthly average departure delay (left) and monthly cancellation rate (right), AA vs DL. Delta consistently posts a lower cancellation rate than American Airlines, while average delay is comparable across both carriers and rises in February and March.

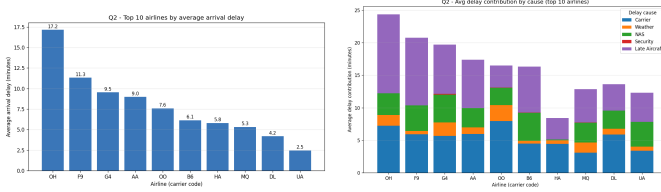


Fig. 3. Q2: top 10 carriers by mean arrival delay (left) and the stacked breakdown of the five delay causes (right). The dominant contributor for the worst-performing carriers is LATE_AIRCRAFT_DELAY, i.e. the cascading effect of aircraft arriving late from a previous leg.

memory and trigger an extra materialization pass without any downstream consumer.

c) Shuffle partitions tuned to the workload: Spark’s default of 200 shuffle partitions is overkill for a working dataset of 2.23 M records on a single machine. The configuration is reduced to 8, which lines up with the host CPU count and removes the scheduling overhead introduced by lots of mostly-empty partitions.

d) Coalesce is a UX/cost trade-off: `coalesce(1)` guarantees a single CSV file per query (the project deliverable wants exactly one per query) at the cost of a trailing shuffle that shows up in the output timer. Dropping `coalesce(1)` would shave roughly 1–3s but would also produce eight `part-*` files that the user would then have to concatenate by hand.

VII. CONCLUSIONS

This project demonstrates a complete, reproducible Big Data pipeline that combines HDFS storage and Spark processing, both deployed via Docker Compose so that the cluster can be brought up from scratch in under a minute. The two required queries are implemented with the DataFrame API, paying explicit attention to lazy evaluation, caching and shuffle behaviour. The experimental evaluation quantifies what it costs to move from a local PySpark setup to a containerized HDFS+Spark cluster, and identifies loading (network and metadata) and output (`coalesce` + HDFS replication) as the dominant overhead sources. The full source code, the result CSVs, the benchmark logs and this report are organized in a single repository as required by the project specification.

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